Tax Heterogeneity and Misallocation*

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Abstract

There is substantial asymmetry in effective corporate income tax rates across firms. While tax asymmetries reduce productivity in frictionless economies, they can improve efficiency in distorted economies by alleviating other frictions. We develop a framework to estimate the productivity effects of tax asymmetries in distorted economies. Using U.S. firm-level balance sheet data alongside measures of marginal tax rates, we find a positive correlation between tax rates and factor productivity, suggesting that tax asymmetries *exacerbate* other economic frictions. Our model implies significant aggregate productivity gains from eliminating tax asymmetries.

*The views expressed here are those of the authors and do not necessarily reflect the views of the Federal Reserve Bank of Cleveland or the Federal Reserve System.

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1 Introduction

In efficient economies, marginal products of inputs are equalized across firms. Yet, numerous studies following Hsieh and Klenow (2009) document substantial productivity differences across firms, suggesting significant gains from resource reallocation. The sources of input misallocation, however, remain elusive, preventing policy guidance.

We examine whether corporate taxation contributes to misallocation. Although the tax code does not distinguish individual firms *de jure*, provisions for deductions and allowances, such as imperfect loss offsets or interest deductions, create *de facto* asymmetries in effective marginal tax rates (EMTRs) across firms. This can cause factor productivity to diverge across firms and reduce efficiency. In our sample of U.S. firms, EMTRs vary widely across firms, with large firms typically facing higher rates (Figure 1). Because these firms represent the majority of economic activity, the efficiency implications of tax asymmetries are potentially large, especially if size indicates productivity.

Whether tax asymmetries reduce efficiency is theoretically ambiguous. Although they impair efficiency in competitive, otherwise frictionless economies, they might actually *improve* efficiency in distorted economies by offsetting existing frictions. For instance, interest deductions might ease credit constraints and raise investment, while loss-offset provisions might prevent premature bankruptcy among liquidity-constrained firms by reducing their tax obligations at times of distress. Whether this is the case is an empirical question.

In pursuit of an answer, we study the effect of tax asymmetries on total factor productivity (TFP) in a production model with heterogeneous firms and distortions to factor allocation. We derive analytical formulas that relate the joint cross-sectional distribution of EMTRs and factor productivities to aggregate TFP. The key insight is that the relationship between corporate taxation and other distortions is captured by the correlation between EMTRs and productivity across firms. A positive correlation implies that high-productivity firms are taxed more heavily, lowering TFP.

Using firm-level data on EMTRs, we estimate that eliminating tax asymmetries would *raise* TFP by around 4%, holding technology and other distortions fixed. These gains arise from the

positive empirical correlation between EMTRs and firm productivity: tax homogenization lowers EMTRs for more productive firms and reallocates capital toward them, thereby raising TFP. Because capital and labor are complementary, this capital reallocation also induces labor reallocation, which further increases TFP given the strong positive correlation between EMTRs and labor productivity. Abstracting from these correlations, the implied productivity gains from tax homogenization are below 1%, with the exception of the pre-1986 period. This indicates that tax asymmetries amplify rather than offset existing misallocation.

We then discuss forces that can modify both the magnitude and the interpretation of the productivity gains from tax homogenization. First, we trace out the dynamic implications of these productivity gains and show that induced capital accumulation and revenue-neutral tax adjustments further amplify long-run output beyond the static TFP improvements. Second, we allow for interactions between tax distortions and other frictions—such as imperfect deductibility of labor costs and financial constraints—and show that these interactions can substantially amplify the productivity gains when all observed correlations are interpreted causally. Third, the results are robust to major macroeconomic developments over our sample period, including the secular decline in the labor share of income. Finally, we find that the TFP improvements arise predominantly from a more efficient resource allocation within sectors, indicating that tax heterogeneity mainly distorts competition among firms in the same industry.

This paper connects to the literature on the macroeconomic effects of factor misallocation (Restuccia and Rogerson, 2008; Hsieh and Klenow, 2009). Papers in this literature typically adopt one of two approaches. The first approach measures the dispersion in marginal products and attributes it to firm-specific distortions. Comparing the distorted economy with a frictionless environment, these papers typically find large gains from removing frictions (see e.g., Hsieh and Klenow (2009), Gopinath, Kalemli-Özcan, Karabarbounis, and Villegas-Sanchez (2017), and Adamopoulos, Brandt, Leight, and Restuccia (2022)). A challenge in this literature has been to distinguish differences in marginal products from differences in production technologies or from measurement

error (Bils, Klenow, and Ruane, 2021).[1]

The second approach measures frictions at the firm level and computes the associated reductions in aggregate productivity (Midrigan and Xu, 2014; David and Venkateswaran, 2019; Kim, 2023; Stefanski, Ahlvik, Andersen, Harding, and Trew, 2025). Because of computational and theoretical complexities, this approach is limited in its ability to model and compute a large number of distortions and their interactions. Because specific frictions only partly explain the differences in firms' marginal products, most of these studies find limited gains from reallocation.

We bridge these approaches by focusing on a specific friction—corporate tax asymmetries—while explicitly allowing for other distortions. In the spirit of Lipsey and Lancaster's (1956) analysis of the 'second best', removing tax heterogeneity in a frictional environment need not increase aggregate TFP, in contrast to most previous work, where eliminating distortions improves productivity by design. Our approach is closer to Harberger (1964), and, more recently, Baqaee and Farhi (2020), Qian and Vereshchagina (2024), and Klenow, Pastén, and Ruane (2025), who respectively study productivity effects of price markups, financial constraints, and carbon taxes in distorted economies.[2] More broadly, our methodology provides a general framework for quantifying the allocative efficiency costs of any measurable distortion that creates factor-cost wedges across firms.

Our paper also relates to the broader corporate taxation literature (Harberger, 1962; Hall and Jorgenson, 1967). This literature has studied specific aspects of the tax code, such as loss-offset provisions (Auerbach, 1986; Kaymak and Schott, 2019), investment tax credits (Cummins, Hassett, and Hubbard, 1994), jurisdictional differences (Djankov, Ganser, McLiesh, Ramalho, and Shleifer, 2010; Slattery and Zidar, 2020), as well as tax avoidance (Glover and Levine, 2023). Whereas these papers focus on the economic effects of the *level* of the corporate tax rate, we study the *dispersion* in tax rates and its implications for aggregate productivity.

[1] Hsieh and Klenow (2009) address these issues by benchmarking their misallocation measure to the U.S. when assessing the extent of resource misallocation in India. Bils et al. (2021) use panel data to purge the firm-level dispersion in productivity of measurement error. Because our findings rely on the covariance between tax wedges and productivity, errors in measurement of marginal products are less of a concern, see Section 3.

[2] In Klenow et al. (2025), a uniform carbon tax differentially affects firms due to differences in fuel use intensity. Implications for aggregate productivity depend on the correlation between fossil fuel use and revenue productivity, which reflects output distortions in the economy.

2 Model

Consider a firm i that uses capital and labor to produce output with a decreasing-returns-to-scale production technology, $y^i = z^i F(k^i, n^i)$. Markets are competitive. A firm's pre-tax profits are given by $\Pi^i_{\text{pre-tax}} = z^i F(k^i, n^i) - \omega^i_N n^i - \omega^i_R k^i$, where ω^i_N denotes the marginal cost of labor and $\omega^i_R = r^i + \delta^i$ the marginal cost of capital, consisting of the real rental rate r^i and the depreciation rate δ^i.[3]

Profits are taxed at firm-specific rates. Although the tax code does not distinguish individual firms *de jure*, effective marginal tax rates can differ across firms. This reflects special provisions that shape the tax base, in particular loss provisions and various deductions and allowances. Loss offsets—which allow firms to carry losses forward or back to offset taxable income in other periods—tie current taxable income to a firm's past performance and give rise to firm-specific income tax rates τ^i_b.

When production expenses are not fully deductible from income, as is the case for capital costs, corporate income taxes can distort factor demand. The effective distortions to capital and labor are determined by the income tax rate τ^i_b and any deductions from taxable income; we illustrate this using depreciation allowances for capital and full expensing of labor costs.[4] Depreciation allowances determine the timing of cost recovery for tax purposes. Let d_{it} denote firm i's tax-depreciation allowance in year t, defined as the fraction of the asset's initial acquisition cost that the tax code permits the firm to deduct from taxable income in that year.[5] Following Hall and Jorgenson (1967), denote the present value of these depreciation allowances as

$$\lambda^i = \sum_{t=0}^{\infty} \frac{d_{it}}{(1+r_i)^t}. \tag{1}$$

[3] In Appendix A, we demonstrate how capital adjustment costs and credit constraints can generate firm-specific marginal costs.

[4] In Section 5, we consider imperfect expensing of labor costs. Appendix A shows a more comprehensive case that also includes debt financing.

[5] Tax-depreciation schedules generally differ from economic depreciation δ. The particular formulation of d_{it} depends on which methods were allowed that year (e.g., straight-line vs. double declining balance), the tax lives of capital types (e.g., structures vs. equipment), or the accounting method used (e.g. whether inventories of the firm follow last-in-first-out or first-in-first-out accounting). See Hall and Jorgenson (1967) and Gravelle (2001) for details.

A higher λ^i corresponds to more front-loaded depreciation for tax purposes and therefore to a larger tax shield per unit of capital. The present value of the tax base associated with the marginal investment is then given by $z^i F(k^i, n^i) - \omega_N^i n^i - \omega_R^i \lambda^i k^i$. The firm's problem consists of choosing k and n to maximize after-tax profits.

$$\max_{k,n} (1 - \tau_b^i) \left[z^i F(k, n) - \omega_N^i n \right] - \omega_R^i \left(1 - \tau_b^i \lambda^i \right) k \tag{2}$$

The optimality condition for labor is

$$z^i F_n(k, n) = \omega_N^i,$$

which equates marginal product of labor to its marginal cost. Because labor costs are fully deductible, corporate taxation distorts employment only indirectly, through the distortion to capital. The optimality condition for capital is

$$z^i F_k(k, n) = \omega_R^i \cdot \underbrace{\frac{1 - \tau_b^i \lambda^i}{1 - \tau_b^i}}_{\omega_\tau^i} \equiv \omega_K^i, \tag{3}$$

where $z^i F_k(k, n)$ is the marginal product of capital and ω_K^i denotes the tax-adjusted marginal cost of capital. The term ω_K^i is composed of a pre-tax component ω_R^i and a tax distortion ω_τ^i, which embeds all the information about the effective rate before deductions τ_b^i and the depreciation allowance λ^i. Equation (3) can equivalently be derived by formulating the firm's problem in (2) using an *effective* corporate income tax rate $\tau^i = 1 - 1/\omega_\tau^i$ that consolidates all deductions—such as loss offsets and depreciation allowances—into a single wedge.[6] Estimates of τ were shown in Figure 1. Throughout, τ^i should be interpreted as the effective marginal tax rate on an additional unit of capital investment, rather than the statutory tax rate.

Equation (3) shows how distortions and tax asymmetries might affect aggregate productivity by

[6]The corresponding effective tax rate is $\tau^i = 1 - (1 - \tau_b^i)/(1 - \tau_b^i \lambda^i)$. Appendix A shows another example, where capital is partly financed with debt.

creating differences in ω_K^i across firms. We focus here on the role of tax asymmetries, i.e., differences in ω_τ^i. That role generally depends on how ω_τ^i interacts with other distortions. For instance, if firms had common values of ω_R^i, then tax asymmetries would distort the capital allocation and reduce TFP. But, if ω_R^i differed across firms and tax rates were $\omega_\tau^i = 1/\omega_R^i$ instead, taxes would offset capital distortions and improve TFP.

To formally define an equilibrium, rewrite $\omega_R^i = (r+\delta)\tilde{\omega}_R^i$ and $\omega_N^i = w\tilde{\omega}_N^i$. Let $s = (z, \tau, \tilde{\omega}_N, \tilde{\omega}_R)$ be the firm's state and $\tilde{G}(s)$ its joint distribution. Define aggregate wedge rents: $X = w\int(\tilde{\omega}_N^i - 1)n^i\, d\tilde{G} + (r+\delta)\int(\tilde{\omega}_R^i - 1)k^i\, d\tilde{G}$. These are the aggregate infra-marginal returns arising from economic distortions.

Definition 1 (Baseline economy)**.** *A* steady-state equilibrium *consists of prices* (r, w)*, quantities* (c, K)*, transfers* T*, dividends* D*, wedge rents* X*, and firm allocations* $\{k(s), n(s)\}$ *such that (i) firms choose* $k(s)$ *and* $n(s)$ *to maximize after-tax profits,* $\pi(s) = \max_{k,n}\ (1-\tau)\big[zF(k,n) - w\tilde{\omega}_N n\big] - (r+\delta)\tilde{\omega}_R\, k$*, (ii) households consume factor income, dividends, and transfers:* $c = wN + rK + D + T + X$*, (iii) factor markets clear:* $\int k(s)\, d\tilde{G}(s) = K$ *and* $\int n(s)\, d\tilde{G}(s) = N$*, (iv) the government budget balances:* $T = \int \tau(s)\big[zF(k(s), n(s)) - \omega_N(s)n(s)\big]\, d\tilde{G}(s)$*, and (v) dividends equal aggregate after-tax profits:* $D = \int \pi(s)\, d\tilde{G}(s)$.

These conditions imply that the resource constraint holds: $c = \int zF(k(s), n(s))\, d\tilde{G}(s) - \delta K$.

2.1 Tax heterogeneity and aggregate productivity

We begin by formulating TFP in the baseline economy and drop the firm subscript i for readability. We set $F(k,n) = zk^\alpha n^\beta$ with $\alpha + \beta = \gamma < 1$, and denote the joint distribution of firm-level TFP, capital and labor distortions by $G(z, \omega_K, \omega_N)$. The following proposition gives productivity in the baseline equilibrium.[7]

[7] Proofs and derivations are in Appendix A.

Proposition 1. *TFP in the baseline economy is*

$$Z = \frac{Y}{K^{\alpha}N^{\beta}} = \frac{\int z^{\frac{1}{1-\gamma}}\omega_N{}^{-\frac{\beta}{1-\gamma}}\omega_K{}^{-\frac{\alpha}{1-\gamma}}dG}{\left[\int z^{\frac{1}{1-\gamma}}\omega_N{}^{-\frac{\beta}{1-\gamma}}\omega_K{}^{-\frac{1-\beta}{1-\gamma}}dG\right]^{\alpha}\left[\int z^{\frac{1}{1-\gamma}}\omega_N{}^{-\frac{1-\alpha}{1-\gamma}}\omega_K{}^{-\frac{\alpha}{1-\gamma}}dG\right]^{\beta}}. \tag{4}$$

Aggregate TFP depends on the joint distribution of firm productivity z, and distortions ω_N and ω_K. The key is the relative distortion across firms rather than the level of distortions: a proportional change in distortions does not affect TFP.

The baseline economy is distorted along multiple dimensions. To focus on the implications of tax distortions alone, we consider a uniform-tax economy where all firms face the same marginal tax rate: $\tau^i = \bar{\tau}$. We leave the distribution of other distortions intact, i.e., all firms continue to face the same distortions to employment but capital distortions are altered to reflect a common tax wedge. We therefore adopt an agnostic view of the sources of differences in ω_R^i and ω_N^i. In Section 5, we consider the possibility that tax policy interacts with the underlying distortions and propose bounds on how those interactions may affect our analysis.

Our aim is to compare aggregate productivity in that uniform-tax economy with the baseline economy. The counterfactual TFP under uniform taxation is obtained by replacing ω_K with $\omega_K' = \omega_R \cdot \bar{\omega}_\tau$ in equation (4), where $\bar{\omega}_\tau = 1/(1-\bar{\tau})$.

Proposition 2. *TFP in the uniform-tax economy with homogeneous tax rates across firms is*

$$Z^* = \frac{Y}{K^{\alpha}N^{\beta}} = \frac{\int z^{\frac{1}{1-\gamma}}\omega_N{}^{-\frac{\beta}{1-\gamma}}\omega_R{}^{-\frac{\alpha}{1-\gamma}}dG}{\left[\int z^{\frac{1}{1-\gamma}}\omega_N{}^{-\frac{\beta}{1-\gamma}}\omega_R{}^{-\frac{1-\beta}{1-\gamma}}dG\right]^{\alpha}\left[\int z^{\frac{1}{1-\gamma}}\omega_N{}^{-\frac{1-\alpha}{1-\gamma}}\omega_R{}^{-\frac{\alpha}{1-\gamma}}dG\right]^{\beta}}. \tag{5}$$

Because proportional changes in distortions do not affect aggregate TFP, $\bar{\omega}_\tau$ is absent in equation (5); the uniform-tax economy is equivalent to an economy where tax wedge is removed altogether: $\omega_k' = \omega_R$.

The ratio Z^*/Z gives the effect of eliminating tax heterogeneity on aggregate productivity. From equation (5), $Z^* = Z$ when ω_τ is constant across firms and $Z^* > Z$ whenever ω_τ is hetero-

geneous but orthogonal to other distortions.[8] More generally, however, if tax rates *correlate* with other distortions, Z^* can be higher or lower than Z.

To gain some intuition, consider an economy where distortions are distributed jointly according to a log-normal density. The resulting formulas demonstrate how interactions between tax rates and other distortions are essential to the allocative effects of tax asymmetries. Under log-normality, TFP in the baseline economy is given by

$$\ln Z = \ln \int z^{\frac{1}{1-\gamma}} dG - \frac{1}{2}\frac{1}{1-\gamma}\left[\alpha(1-\beta)\sigma_k^2 + \beta(1-\alpha)\sigma_n^2 + 2\alpha\beta\sigma_{kn}\right], \tag{6}$$

where σ_k^2 and σ_n^2 denote the variances of $\ln \omega_K$ and $\ln \omega_N$, and σ_{kn} is their covariance. Aggregate TFP reflects the underlying distribution of firm productivity, z, adjusted for efficiency losses from input distortions. We are interested in the change in TFP from eliminating tax heterogeneity. This implies setting $\sigma_k^2 = \sigma_R^2$ and $\sigma_{kn} = \sigma_{Rn}$ in equation (6). Taking differences gives the next proposition.

Proposition 3. *If ω_K, ω_N, and ω_τ are jointly log-normally distributed, then eliminating tax heterogeneity yields the following change in aggregate TFP:*

$$\ln \frac{Z^*}{Z} = \frac{1}{2}\frac{\alpha(1-\beta)}{1-\gamma}\sigma_\tau^2 + \frac{\alpha(1-\beta)}{1-\gamma}\sigma_\tau^2(L_{k\tau} - 1) + \frac{\alpha\beta}{1-\gamma}\sigma_\tau^2 L_{n\tau}, \tag{7}$$

where σ_τ^2 is the variance of $\ln \omega_\tau$ and $L_{k\tau} = \sigma_{k\tau}/\sigma_\tau^2$ and $L_{n\tau} = \sigma_{n\tau}/\sigma_\tau^2$ denote the slope coefficients from a linear projection of $\ln \omega_K$ and $\ln \omega_N$ on $\ln \omega_\tau$.

Equation (7) has three components. The first one is a pure misallocation component, representing the reduction in aggregate output caused by tax asymmetries. In the absence of other distortions, or if such distortions were orthogonal to tax rates, this would be the expected TFP improvement from equalizing EMTRs. Productivity gains are increasing in the variance of tax wedges, σ_τ^2, the returns to scale γ, and the capital share α.[9]

[8] This follows from Jensen's inequality combined with the fact that $1 - \beta > \alpha$.

[9] To see this, note that $\frac{\alpha(1-\beta)}{1-\gamma} = \alpha + \frac{\alpha^2}{1-\gamma}$.

The second term captures the correlation between taxes and other capital distortions. If tax differences alleviate other capital distortions, then $\ln \omega_R$ and $\ln \omega_\tau$ are negatively correlated, and $L_{k\tau} < 1$. Consequently, eliminating tax differentials lowers TFP. Taxes could also exacerbate other distortions. If firms with higher capital costs also face higher tax rates, then $L_{k\tau} > 1$, which raises the gains from eliminating tax differentials.

The last term captures the correlation of taxes with labor distortions. Tax homogenization lowers the tax rates for some firms and raises their capital demand. Because capital and labor are complements in production, labor gets reallocated toward those firms. This improves efficiency only if the marginal labor productivity is higher at those firms on average ($L_{n\tau} > 0$).

In our quantitative analysis, we implement equations (4) and (5) to estimate the productivity gains from uniform taxation and use equation (7) to understand its sources.

3 Data and methodology

Our main data source is the annual Compustat database from 1980–2021, which provides balance-sheet data on publicly listed companies (Standard & Poor's, 2023). Our formulas require information on employment, capital, and output. For each firm-year, we define output as the sum of sales and changes in inventories. We construct a firm's capital stock from its investment expenditures using the perpetual inventory method.[10]

3.1 Measuring tax distortions

We merge the Compustat data with estimates of corporate income tax rates for each firm-year from Graham and Mills (2008) and from Blouin, Core, and Guay (2010). These studies simulate expected EMTRs at the firm-year level, τ_{it}, based on the corporate income tax code in effect during that year. They allow for provisions such as investment tax credits, the alternative minimum tax,

[10] See Appendix B for details.

capital expensing rules, and tax carryovers, and measure the EMTRs after interest deductions.[11] This information is combined with company-level balance sheet data from Compustat, e.g., detailed capital expenditures and past tax carryovers. Each study estimates a stochastic process for a company's income to predict the applicable tax rates on future income associated with that year's investment and reports the average marginal tax rate across possible income realizations (Graham, 1996a,b). The simulated rates reflect the expected marginal tax rate on investment. Graham and Mills (2008) show that the simulated taxes closely approximate the actual taxes from tax records. These simulated marginal tax rates are the empirical counterparts to τ^i in the model. The income-process assumptions used to construct these rates affect the measurement of EMTRs but are not structural assumptions of our model, which takes the resulting rates as inputs and asks how their cross-sectional dispersion affects factor allocation and aggregate TFP.

Both measures are well below the statutory rates and vary significantly across firms. The two measures, denoted τ^{GM} and τ^{BCG}, average 17 and 24 percent with standard deviations of 17 and 14 percent, mostly attributable to cross-sectional differences (Table A1). While the two simulated rates are different, they are highly correlated ($\rho = 0.62$).

We deploy both measures in our analysis below by modeling them as noisy measures of the same latent EMTR:

$$\ln \omega^*_{j\tau} = \ln \omega_\tau + \epsilon_j,$$

where ϵ_j is a classical measurement error with variance $\sigma^2_{j\epsilon}$ and conditional mean $\mathbb{E}[\epsilon_j|\omega_\tau] = 0$ for $j \in \{\text{GM}, \text{BCG}\}$. We assume orthogonality between the two measurement errors, $\text{Cov}\left(\epsilon_{\text{GM}}, \epsilon_{\text{BCG}}\right) = 0$. We view this as plausible because the measures differ primarily in their income-process assumptions: Graham and Mills (2008) impose a random walk, whereas Blouin et al. (2010) allow for mean reversion. The resulting forecasts of conditional income volatility affect the probability that loss carryforwards and other tax shields are exhausted and hence affect the simulated EMTRs.

Measurement error in tax wedges exaggerates tax asymmetries and attenuates the estimated

[11] Appendix A.2 shows how an after-interest-deduction EMTR maps into the model's tax wedge when investment is partly debt-financed.

relationship between tax wedges and capital and labor productivity. These biases lead to underestimation of the TFP gain, as the next proposition states, and distort the assessment of its components in equation (7).

Proposition 4. *Assume that the tax wedge is measured with error,* $\ln \omega_\tau^* = \ln \omega_\tau + \epsilon$ *with* $\mathbb{E}(\epsilon|\omega_\tau) = 0$. *Then, replacing* ω_τ *by* ω_τ^* *in equation* (7) *underestimates the net TFP gain from eliminating tax heterogeneity:*

$$\ln(Z^*/Z)|\omega_\tau^* = \ln(Z^*/Z)|\omega_\tau - \frac{\sigma_\epsilon^2}{2}\frac{\alpha(1-\beta)}{1-\gamma} \tag{8}$$

We compute TFP gains using the general formulas in equations (4) and (5), grouping firms into tax-rate quantiles within each year. As shown in the Appendix, equation (8) remains applicable when marginal products are conditionally log-normal even if tax wedges are not. To correct for the downward bias in the estimated TFP gain due to measurement error, we estimate the variance of the tax wedge as $\hat{\sigma}_\tau^2 = \mathrm{Cov}(\ln \omega_{\mathrm{GM},\tau}^*, \ln \omega_{\mathrm{BCG},\tau}^*)$ and each measure's error variance as $\hat{\sigma}_{j\epsilon,t}^2 = \mathrm{Var}(\ln \omega_{j\tau}^*) - \hat{\sigma}_\tau^2$ for each year.

Estimating TFP gains and decomposing them into their components using equation (7) requires consistent estimates of $L_{k\tau}$ and $L_{n\tau}$, which we turn to next.

3.2 Correlations between productivity and tax distortions

To measure capital and labor distortions, we use the optimality conditions for factor demand, $\ln \omega_K = \ln \alpha + \ln(y/k)$ and $\ln \omega_N = \ln \beta + \ln(y/n)$, where factor shares α and β are assumed common to all firms in an industry during a given year. Specifically, we estimate

$$\ln(y/k)_{it} = D_{st}^k + L_{k\tau,t} \ln \omega_{\tau,it}^* + e_{it}^k, \tag{9}$$

where i denotes the firm, and t the year. D_{st}^k are indicators for a full set of sector and year interactions. These indicators capture variations in capital shares and average distortions across sectors and years. Similarly, we estimate the cross-sectional correlation between labor distortions and tax

wedges by:

$$\ln(y/n)_{it} = D^n_{st} + L_{n\tau,t} \ln \omega^*_{\tau,it} + e^n_{it}, \tag{10}$$

Two remarks are in order. First, the traditional endogeneity concern familiar from production function estimation does not apply here since these specifications replace inputs with cost parameters, ω_K or ω_N, using the optimality conditions. Second, the goal is to recover projection coefficients that implicitly describe how the tax wedge covaries with (unobserved) factor-cost wedges in the cross section, not to estimate the causal effect of tax rates on firm outcomes. $L_{k\tau,t}$ and $L_{n\tau,t}$ summarize the covariances of the tax wedge with the total capital and labor wedges, which are the sufficient statistics for whether tax asymmetries amplify or offset other sources of misallocation.[12] The deeper assumption behind our baseline approach is the exogeneity of non-tax cost wedges to tax policy, which may be violated in some models, e.g., those with collateral constraints. We revisit this issue in Section 5.

A separate concern is measurement error in the tax wedge, which attenuates OLS estimates of $L_{k\tau}$ and $L_{n\tau}$ toward zero. We therefore estimate equations (9) and (10) with an IV approach, where one tax measure is used as an instrument for the other. This yields consistent estimates when the measurement errors in the two tax measures are orthogonal to each other and to any component of the residual that does not reflect a factor-cost wedge, such as measurement error in y, k, or n, or specification error from deviations of marginal products from average products.

Any measurement error common to both tax estimates, e.g., from reporting errors in Compustat or in the estimated earnings processes, would violate the first condition. If these common errors are also correlated with residual factor productivity—for example, through mismeasurement of output or inputs—the direction of the IV bias is ambiguous. If instead they are orthogonal to residual factor productivity, they bias the IV estimates toward the OLS estimates. Because we find larger IV than OLS estimates, the true losses from tax misallocation would then be larger than documented here.

To summarize the correlation patterns between different distortions, we report pooled estimates

[12] Recall that $L_{k\tau} = \sigma_{k\tau}/\sigma^2_\tau = 1 + \sigma_{R\tau}/\sigma^2_\tau$ and $L_{n\tau} = \sigma_{n\tau}/\sigma^2_\tau$.

for all years in Table 1. Columns 1-3 show the OLS estimates of (9) and (10). Panel A uses tax measures from Graham and Mills (2008), and Panel B uses the tax measures from Blouin et al. (2010). Both measures suggest a strong positive relationship between taxes and the three productivity measures, although they somewhat disagree on the strength of that relationship.

The implied correlation between tax wedges and other capital distortions (Column 1) differs across the two measures. Recall from (7) that a value below (above) one indicates that the tax wedge is negatively (positively) correlated with other capital distortions. While Graham and Mills's EMTR estimates suggest tax wedges and capital distortions correlate negatively, Blouin et al.'s estimates suggest they are orthogonal. By contrast, because both coefficients are larger than zero, the two measures suggest a positive correlation between labor distortions and the tax wedge (Column 2). This implies that reallocating labor toward high-tax firms would raise productivity. Column 3 shows the projections of firm-level TFP on the tax wedge, which we use below for some of our results. Consistent with the patterns from average labor and capital productivity, these estimates show that productive firms on average face higher marginal tax rates.

Correcting for attenuation bias depicts a more consistent picture across measures (Columns 4-6). Both tax measures now indicate that tax wedges correlate positively with other capital distortions, implying that tax asymmetries exacerbate the existing capital distortions.

With two tax measures, we have two estimates of the same underlying parameter for each year. In our calculations of TFP, we combine them by taking an average, weighted by the inverse of their variance. For example, using the estimates for the entire sample period, reported in columns 4 and 5 of Table 1, we obtain a value of 1.41 (s.e. 0.06) for $L_{k\tau}$ and 1.48 (s.e. 0.04) for $L_{n\tau}$ for the entire sample period (see Appendix B).

Before we turn to the implications of our estimates for aggregate TFP, we need to set values for the parameters of the production function. For our baseline results, we keep those parameters fixed over time. This allows us to highlight the changes in the interactions between distortions when presenting the time trends. We set $\gamma = 0.85$, which implies a profit share of 0.15 in output, roughly the long-run average value of the profit share in Barkai and Benzell (2024). We set $\beta = \gamma \times 2/3$,

which gives a labor share of 0.57, in line with the BEA's (0.55) and the BLS's (0.60) estimates for our sample period. The resulting capital share is $\alpha = \gamma - \beta = 0.28$. Section 5 shows that alternative calibrations that allow input shares to vary across sectors or over time yield similar results.

4 Results

Our primary finding is that eliminating tax heterogeneity would improve U.S. TFP. Our baseline estimates put this gain between 4 and 5 percent. When we decompose the potential TFP gains from eliminating tax heterogeneity into three components, we find that the positive correlation of tax rates with capital and especially labor distortions accounts for the largest share of the TFP gains. Tax asymmetries alone explain a sizable part of the estimated TFP gains, but this component became smaller after the tax reforms of the 1980s.

Figure 2 shows the estimated TFP gains associated with eliminating tax heterogeneity. The solid black line shows our baseline result. The gain is positive throughout our sample period because, on average, factors are more productive at firms that face higher marginal tax rates. This suggests that tax heterogeneity exacerbates capital and labor distortions. The TFP gain ranges from 4 to 5 percent during the early 1980s to around 3 percent in recent years. The average gain across all years is 4.1%, which corresponds to four years of average annual U.S. TFP growth during our sample period.

To understand the sources of TFP gains, we use our approximation formula in equation (7). The gray line in Figure 2 shows the approximate gains with the 95% confidence band shaded in gray. Using the approximation, the average TFP gain is 4.6%, close to the 4.1% benchmark. This reinforces our main finding that tax heterogeneity reduces aggregate TFP. The two series are highly correlated ($\rho = 0.96$), lending credence to our decomposition results, which we turn to next.

The three components of the TFP gains from equation (7) are shown in Figure 3. The total gains are shown as the solid gray line in Figure 3, which is identical to the gray line in Figure 2.

The blue bars reflect misallocation from pure tax heterogeneity. If tax wedges were uncorrelated with the marginal products of capital and labor, the blue bars would represent the TFP gain from tax asymmetries in an otherwise frictionless economy. This component was 1-2% prior to 1986, due to the higher statutory corporate tax rate, and has been under 1% since then, amounting to less than a quarter of the total TFP gain.

The red bars represent the TFP gains due to the correlation between tax wedges and capital distortions. In most years, tax rates have a weakly positive correlation with capital distortions, resulting in additional TFP gains of less than 1%. The potential gains implied by this component have diminished over time, turning negative in recent years. This suggests an efficiency improvement from recent changes in the distribution of tax rates.

The yellow bars represent the TFP gain associated with the correlation between labor distortions and tax wedges. When that correlation is positive, equalizing taxes reduces rates for firms with higher marginal product of labor. Because of capital-labor complementarity, the ensuing employment reallocation toward those firms raises TFP. Quantitatively, this is the largest component of the TFP gain, partly because labor's share of income is roughly twice the capital share.[13]

Overall, the majority of the estimated TFP gain is attributable to the positive correlation between tax wedges and labor distortions. This highlights the importance of taking other distortions into account when studying the allocative effects of a particular distortion.

5 Robustness, extensions and long-run implications

In this section, we explore the robustness of our findings, examine how they change under alternative modeling assumptions, and discuss their implications for long-run output. Table 2 summarizes our results. Online Appendix A provides the detailed methodology for different modeling extensions and presents the full results for each of the subsections below.

[13] More generally, capital and labor co-move during the reallocation from eliminating tax heterogeneity as long as the elasticity of substitution between capital and labor is not too high (less than $1/(1-\gamma)$).

5.1 Long-run output effects

The productivity gains from tax homogenization translate into larger long-run output effects through capital accumulation. In a standard growth model with inelastic labor supply, our baseline 4.1% TFP gain implies a 5.8% increase in long-run output per capita, as higher productivity encourages additional investment. Models incorporating elastic labor supply would predict even larger effects.

A related consequence of output growth is higher tax revenue. Revenue-neutral tax reforms call for a lower tax rate, which further encourages capital accumulation. This generates a fiscal multiplier effect: productivity-driven output growth increases tax revenue, permitting lower tax rates that further stimulate capital formation. Accounting for this feedback loop, we estimate revenue-neutral tax homogenization would raise long-run output per capita by approximately 6.4%.[14]

5.2 Interactions between distortions

The baseline model treats the distribution of non-tax distortions, ω_R and ω_N, as exogenous to tax policy. When corporate taxation interacts with other economic frictions, the welfare effects of tax reform may exceed our baseline estimates.

We consider two channels through which such interactions could amplify productivity gains. First, if labor costs are imperfectly deductible—whether due to incomplete expensing provisions or cash-in-advance requirements—corporate tax heterogeneity directly distorts employment decisions beyond its effects on capital allocation. Second, if the observed correlation between tax rates and capital costs reflects a causal relationship (for instance, through collateral constraints that bind more tightly at higher tax rates), then equalizing tax rates would simultaneously reduce dispersion in capital costs.

The results are shown under the heading 'Structural Distortions' in Table 2. Under the extreme assumption that all observed correlations between taxes and factor costs are structural, we estimate that productivity gains could reach an upper bound of 10.7 percent if tax homogeniza-

[14]The model used here abstracts from firm entry and exit. In related work, Kaymak and Schott (2019) show that improved firm-selection in response to tax homogenization can materially amplify its aggregate output effects in the longer run.

tion mitigates underlying labor distortions and 5.7 percent if it mitigates capital distortions (see Online Appendix A.2 for the theoretical framework and calculations). We interpret these scenarios as providing upper bounds on potential gains, as the true extent of such interactions remains model-specific and empirically ambiguous.

5.3 Robustness to macroeconomic trends

Our baseline estimate uses stable factor income shares. The labor share of income declined from 57.7% to 53.0% between 1980 and 2016, raising questions about whether this trend affects our findings. In Online Appendix A.3, we evaluate two scenarios: one attributing the labor share decline entirely to rising capital intensity, and another attributing it to higher markups. The results are summarized in the last two rows of Table 2.

Under rising capital shares, TFP gains remain nearly identical to the baseline, as the increased importance of capital in production roughly offsets compositional effects. Under rising markups, the estimated gains decline modestly—by approximately one percentage point in recent years—because higher markups move the economy away from the perfectly competitive economy where reallocation gains are maximized.[15] Neither scenario substantially alters our central conclusions about the magnitude of productivity losses from tax heterogeneity.

5.4 Sectoral decomposition of productivity gains

Our baseline model estimates aggregate productivity gains using a common, economy-wide labor share parameter for all sectors. In Online Appendix A.4, we re-estimate TFP gains allowing for sector-specific factor elasticities that reflect technological differences across industries. This refinement yields a total TFP gain of 4.3%, similar to our benchmark estimate of 4.1%, confirming that our findings are robust to technological heterogeneity across sectors. Decomposing the sources of these gains reveals that the vast majority—approximately 3.3 percentage points, or 77%

[15] This exercise does not model heterogeneous markups or strategic pricing. If markup dispersion were systematically related to EMTRs, it would constitute an additional source of misallocation beyond our common-γ baseline.

of the total effect—derives from an improved input allocation within sectors. The reallocation of inputs between sectors contributes an additional 1.0 percentage point to aggregate productivity.

6 Conclusion

Our findings show that policies that reduce corporate tax asymmetries would generate significant gains in aggregate productivity and output. The majority of these gains are attributable to the correlation between labor productivity and marginal tax rates: firms that face higher tax rates are typically those where labor is more productive. This highlights the importance of accounting for other frictions when studying the implications of a specific distortion to aggregate efficiency.

In practice, reforms that reduce tax asymmetries can take one of two forms. The first eliminates the provisions that generate firm-specific rates, replacing them with a uniform flat tax on capital income—for example, adopting full loss-offset would remove the distortions created by current loss provisions. A simpler alternative would shift the tax base from income to revenue (or value-added), equalizing the effective tax treatment of both capital and labor across firms. Because a revenue-based tax applies uniformly regardless of financing structure, loss history, or investment timing, it equalizes EMTRs directly—without requiring a complex decomposition of which specific provisions cause the tax asymmetries.

Note, however, that uniform taxation is not necessarily productivity maximizing in distorted economies. Tax designs that seek to maximize aggregate productivity would aim to offset other distortions through firm-specific taxes. The efficient design of corporate taxation is a promising avenue for future research.

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A Theoretical Appendix

This appendix details the derivations in the text and gives the formal proofs for the propositions.

A.1 Reformulating the model with an effective tax rate

The expression for the tax wedge ω_τ^i in (3) can also be derived from a setup in which the firm faces a *pre-tax* cost of capital of ω_R^i and a single effective tax rate τ^i on operating surplus, which includes loss offsets, as well as depreciation allowances:

$$\max_{k,n} (1-\tau^i)\left[zF(k,n) - \omega_N^i n\right] - \omega_R^i k, \tag{A1}$$

The first-order condition of the firm problem in (A1) is

$$zF_k(k,n) = \omega_R^i \cdot \frac{1}{1-\tau^i}, \tag{A2}$$

where $\omega_R^i = r^i + \delta^i$ are pre-tax user-costs. The conditions in (A2) and (3) are equivalent if we define

$$\tau^i = 1 - \frac{1-\tau_b^i}{1-\tau_b^i\lambda^i}. \tag{A3}$$

A.2 Effective tax rate with debt financing

We now consider a more general problem where a fraction of the marginal investment is financed with debt. A fraction ϕ^i of the marginal capital investment is debt-financed, while the remaining fraction $1-\phi^i$ is equity-financed (Auerbach, 1983). In that case, the first-order condition of the firm problem in (2) with respect to capital is:

$$zF_k(k,n) = \omega_R^i \cdot \underbrace{\frac{1-\tau_b^i\lambda^i + \phi\left[i^i(1-\tau_b^i) - r^i\right]}{1-\tau_b^i}}_{\omega_\tau^i}, \tag{A4}$$

where r^i is the after-tax return required by shareholders, and i^i is the interest rate paid to debt holders. With $\phi^i = 0$ we obtain the cost of financing capital under all-equity finance (as in (3)); with $\phi > 0$, the second term in the numerator adjusts the baseline cost by the difference between the after-tax cost of debt and the required return on equity.

As in Appendix A.1, we can derive an implied effective tax rate

$$\tau^i = 1 - (1 - \tau_b^i)\frac{r^i + \delta^i}{(r^i + \delta^i)(1 - \tau_b^i\lambda^i) + \phi^i\left[i(1 - \tau_b^i) - r^i\right]}, \tag{A5}$$

which captures all the information about the loss offsets, depreciation allowances λ^i, and the financing mix. This lets us write the problem as in (A1).

A.3 An investment problem with multiple distortions

To fix ideas, we develop an investment problem with multiple distortions. For notational simplicity, we model only the effective (post-deduction) marginal tax rate τ, as derived in Appendices A.1 and A.2. Consider the investment problem of a firm that uses capital and labor to produce output with the decreasing-returns-to-scale (DRS) production function, $y = zF(k, n)$, using technology z. Each period, the firm chooses capital investment and labor to maximize the payout to shareholders, which depends on current and expected after-tax profits. For simplicity, assume that the firm knows the realization of next period's productivity z and taxes τ at the time of the investment decision. The firm's problem is given by

$$\begin{aligned} V(z,k) &= \max_{n',k',i} \; -i - \Phi\left(\frac{i}{k}\right)k \\ &+ \beta\left[(1-\tau)\left[zF(k',n') - \omega_N n'\right] + E_{z'|z}V(z',k')\right] \end{aligned} \tag{A6}$$

subject to the law of motion for capital and a collateral constraint

$$k' = i + (1-\delta)k \tag{A7}$$

$$i \leq \zeta q k, \tag{A8}$$

with associated multipliers q and μ. The first-order condition for labor is

$$zF_n(k,n) = \omega_N, \tag{A9}$$

where ω_N denotes the effective user cost of labor. The optimality conditions for investment and future capital in the firm's problem in (A6) are given by the first-order condition with respect to investment,

$$i: \quad -1 - \Phi'\left(\frac{i}{k}\right) + q - \mu = 0 \quad \Leftrightarrow \quad q = 1 + \Phi'\left(\frac{i}{k}\right) + \mu \tag{A10}$$

and the first-order condition with respect to future capital,

$$k': \quad \beta\left[(1-\tau)zF_k(k',n') + E_{z'|z}V_k(z',k')\right] - q = 0, \tag{A11}$$

with the envelope condition

$$V_k(z,k) = -\Phi\left(\frac{i}{k}\right) + \Phi'\left(\frac{i}{k}\right)\frac{i}{k} + q(1-\delta) + \mu\zeta q. \tag{A12}$$

This implies the following optimality condition for the choice of capital:

$$q = \beta\left[(1-\tau)zF_k(k',n') - \Phi\left(\frac{i'}{k'}\right) + \Phi'\left(\frac{i'}{k'}\right)\frac{i'}{k'} + q'(1-\delta) + \mu'\zeta' q'\right] \tag{A13}$$

The marginal cost of capital, q, is balanced against the discounted future benefit of an additional unit of capital. This benefit consists of the marginal product of capital (net of taxes), the value of non-depreciated capital, and the effects on the marginal costs of investment as well as on the

financial constraint. To simplify (A13), define the tax wedge as $\omega_\tau \equiv 1/(1-\tau)$, and the remainder as

$$\omega_R \equiv \frac{q}{\beta} - q'(1-\delta) - \mu'\zeta'q' + E_{z'|z}\left[\Phi\left(\frac{i'}{k'}\right) - \Phi'\left(\frac{i'}{k'}\right)\frac{i'}{k'}\right]. \tag{A14}$$

Using the definitions above, we can write (A13) as

$$zF_k(k,n) \quad = \quad \omega_K \quad \equiv \quad \omega_\tau \cdot \omega_R, \tag{A15}$$

where ω_K denotes the effective user cost of capital, comprising tax costs, ω_τ, and costs of capital adjustment or financial constraints, summarized by ω_R. Any additional frictions a firm might be facing would be included in ω_R.

A.4 Proofs and derivations

Proposition 1. *TFP in the baseline economy is*

$$Z = \frac{Y}{K^\alpha N^\beta} = \frac{\int z^{\frac{1}{1-\gamma}}\omega_N{}^{-\frac{\beta}{1-\gamma}}\omega_K{}^{-\frac{\alpha}{1-\gamma}}dG}{\left[\int z^{\frac{1}{1-\gamma}}\omega_N{}^{-\frac{\beta}{1-\gamma}}\omega_K{}^{-\frac{1-\beta}{1-\gamma}}dG\right]^\alpha \left[\int z^{\frac{1}{1-\gamma}}\omega_N{}^{-\frac{1-\alpha}{1-\gamma}}\omega_K{}^{-\frac{\alpha}{1-\gamma}}dG\right]^\beta}. \tag{4}$$

Proof. The profit-maximizing levels of capital, labor, and output are:

$$n = z^{\frac{1}{1-\gamma}}\left(\frac{\beta}{\omega_N}\right)^{\frac{1-\alpha}{1-\gamma}}\left(\frac{\alpha}{\omega_K}\right)^{\frac{\alpha}{1-\gamma}} \tag{A16}$$

$$k = z^{\frac{1}{1-\gamma}}\left(\frac{\beta}{\omega_N}\right)^{\frac{\beta}{1-\gamma}}\left(\frac{\alpha}{\omega_K}\right)^{\frac{1-\beta}{1-\gamma}} \tag{A17}$$

$$y = z^{\frac{1}{1-\gamma}}\left(\frac{\beta}{\omega_N}\right)^{\frac{\beta}{1-\gamma}}\left(\frac{\alpha}{\omega_K}\right)^{\frac{\alpha}{1-\gamma}} \tag{A18}$$

Substituting in the definition of TFP gives the following:

$$Z = \frac{Y}{K^{\alpha}N^{\beta}} = \frac{\int z^{\frac{1}{1-\gamma}}\omega_N{}^{-\frac{\beta}{1-\gamma}}\omega_K{}^{-\frac{\alpha}{1-\gamma}}dG}{\left[\int z^{\frac{1}{1-\gamma}}\omega_N{}^{-\frac{\beta}{1-\gamma}}\omega_K{}^{-\frac{1-\beta}{1-\gamma}}dG\right]^{\alpha}\left[\int z^{\frac{1}{1-\gamma}}\omega_N{}^{-\frac{1-\alpha}{1-\gamma}}\omega_K{}^{-\frac{\alpha}{1-\gamma}}dG\right]^{\beta}}$$

□

Proposition 2. *TFP in the uniform-tax economy with homogeneous tax rates across firms is*

$$Z^* = \frac{Y}{K^{\alpha}N^{\beta}} = \frac{\int z^{\frac{1}{1-\gamma}}\omega_N{}^{-\frac{\beta}{1-\gamma}}\omega_R{}^{-\frac{\alpha}{1-\gamma}}dG}{\left[\int z^{\frac{1}{1-\gamma}}\omega_N{}^{-\frac{\beta}{1-\gamma}}\omega_R{}^{-\frac{1-\beta}{1-\gamma}}dG\right]^{\alpha}\left[\int z^{\frac{1}{1-\gamma}}\omega_N{}^{-\frac{1-\alpha}{1-\gamma}}\omega_R{}^{-\frac{\alpha}{1-\gamma}}dG\right]^{\beta}}. \tag{5}$$

Proof. In the uniform-tax economy, ${\omega_K}' = \bar{\omega}^K\omega_R = \bar{\omega}^K\omega_K/\omega_\tau$ and ${\omega_N}' = \bar{\omega}_n\omega_N$. Proof follows from substituting these in equation (4). The common components of distortions, $\bar{\omega}_n$ and $\bar{\omega}^K$, cancel each other out. □

Proposition 3. *If ω_K, ω_N, and ω_τ are jointly log-normally distributed, then eliminating tax heterogeneity yields the following change in aggregate TFP:*

$$\ln\frac{Z^*}{Z} = \frac{1}{2}\frac{\alpha(1-\beta)}{1-\gamma}\sigma_\tau^2 + \frac{\alpha(1-\beta)}{1-\gamma}\sigma_\tau^2(L_{k\tau}-1) + \frac{\alpha\beta}{1-\gamma}\sigma_\tau^2 L_{n\tau}, \tag{7}$$

where σ_τ^2 is the variance of $\ln\omega_\tau$ and $L_{k\tau} = \sigma_{k\tau}/\sigma_\tau^2$ and $L_{n\tau} = \sigma_{n\tau}/\sigma_\tau^2$ denote the slope coefficients from a linear projection of $\ln\omega_K$ and $\ln\omega_N$ on $\ln\omega_\tau$.

Proof. Let $\mu_x = \mathbb{E}[\ln x]$ and $\sigma_x^2 = \mathbb{V}[\ln x]$ be the mean and variance of the log of a variable x. Under joint log-normality:

$$\begin{aligned}\ln\int y \;=\; & \frac{\alpha}{1-\gamma}\ln\alpha + \frac{\beta}{1-\gamma}\ln\beta + \frac{1}{1-\gamma}(\mu_z - \alpha\mu_k - \beta\mu_n) \\ & + \frac{1}{2}\frac{1}{(1-\gamma)^2}(\sigma_z^2 + \alpha^2\sigma_k^2 + \beta^2\sigma_n^2 - 2\alpha\sigma_{zk} - 2\beta\sigma_{zn} + 2\alpha\beta\sigma_{kn})\end{aligned}$$

$$\begin{aligned}\ln \int k &= \frac{(1-\beta)}{1-\gamma}\ln\alpha + \frac{\beta}{1-\gamma}\ln\beta + \frac{1}{1-\gamma}(\mu_z - (1-\beta)\mu_k - \beta\mu_n) \\ &+\frac{1}{2}\frac{1}{(1-\gamma)^2}(\sigma_z^2 + (1-\beta)^2\sigma_k^2 + \beta^2\sigma_n^2 - 2(1-\beta)\sigma_{zk} - 2\beta\sigma_{zn} + 2(1-\beta)\beta\sigma_{kn})\end{aligned}$$

$$\begin{aligned}\ln \int n &= \frac{\alpha}{1-\gamma}\ln\alpha + \frac{(1-\alpha)}{1-\gamma}\ln\beta + \frac{1}{1-\gamma}(\mu_z - \alpha\mu_k - (1-\alpha)\mu_n) \\ &+\frac{1}{2}\frac{1}{(1-\gamma)^2}(\sigma_z^2 + \alpha^2\sigma_k^2 + (1-\alpha)^2\sigma_n^2 - 2\alpha\sigma_{zk} - 2(1-\alpha)\sigma_{zn} + 2\alpha(1-\alpha)\sigma_{kn})\end{aligned}$$

Using these equations, the TFP in the distorted economy is:

$$\ln Z = \mu_z + \frac{1}{2}\frac{1}{1-\gamma}\left[\sigma_z^2 - \alpha(1-\beta)\sigma_k^2 - \beta(1-\alpha)\sigma_n^2 - 2\alpha\beta\sigma_{kn}\right]. \tag{A19}$$

With uniform taxation, capital distortions are simply ω_R, which gives the TFP as:

$$\ln Z^* = \mu_z + \frac{1}{2}\frac{1}{1-\gamma}\left[\sigma_z^2 - \alpha(1-\beta)\sigma_R^2 - \beta(1-\alpha)\sigma_n^2 - 2\alpha\beta\sigma_{Rn}\right]. \tag{A20}$$

Note that $\sigma_k^2 - \sigma_R^2 = \sigma_\tau^2 + 2\sigma_{R\tau}$ and $\sigma_{kn} - \sigma_{Rn} = \sigma_{n\tau}$. Rearranging the terms and netting out μ_z, σ_z^2 and σ_n^2 terms, the efficiency losses from distortions are equivalent to:

$$\begin{aligned}\ln\frac{Z^*}{Z} &= \frac{1}{2}\frac{1}{1-\gamma}\left[\alpha(1-\beta)(\sigma_k^2 - \sigma_R^2) + 2\alpha\beta(\sigma_{kn} - \sigma_{Rn})\right] \\ &= \frac{1}{2}\frac{1}{1-\gamma}\left[\alpha(1-\beta)(\sigma_\tau^2 + 2\sigma_{R\tau}) + 2\alpha\beta\sigma_{n\tau}\right] \\ &= \frac{\sigma_\tau^2}{2}\frac{1}{1-\gamma}\left[\alpha(1-\beta)(1 + 2\frac{\sigma_{R\tau}}{\sigma_\tau^2}) + 2\alpha\beta\frac{\sigma_{n\tau}}{\sigma_\tau^2}\right] \\ &= \frac{\sigma_\tau^2}{2}\frac{1}{1-\gamma}\left[\alpha(1-\beta)(1 + 2(L_{k\tau} - 1)) + 2\alpha\beta L_{n\tau}\right] \\ &= \frac{1}{2}\frac{1}{1-\gamma}\alpha(1-\beta)\sigma_\tau^2 + \frac{\alpha(1-\beta)}{1-\gamma}\sigma_\tau^2(L_{k\tau} - 1) + \frac{\alpha\beta}{1-\gamma}\sigma_\tau^2 L_{n\tau}\end{aligned}$$

The last two equalities substitute the linear projection coefficients for $\frac{\sigma_{R\tau}}{\sigma^2_\tau} = L_{R\tau} = L_{k\tau} - 1$ and $\frac{\sigma_{n\tau}}{\sigma^2_\tau} = L_{n\tau}$. □

Proposition 4. *Assume that the tax wedge is measured with error,* $\ln \omega^*_\tau = \ln \omega_\tau + \epsilon$ *with* $\mathbb{E}(\epsilon|\omega_\tau) = 0$*. Then, replacing* ω_τ *by* ω^*_τ *in equation* (7) *underestimates the net TFP gain from eliminating tax heterogeneity:*

$$\ln(Z^*/Z)|\omega^*_\tau = \ln(Z^*/Z)|\omega_\tau - \frac{\sigma^2_\epsilon}{2}\frac{\alpha(1-\beta)}{1-\gamma} \tag{8}$$

Proof. Let $\sigma^2_{\tau^*} = \sigma^2_\tau + \sigma^2_\epsilon$ denote the variance of the measured log tax wedge. Define the projection $\ln x = L_{x\tau} \times \ln \omega_\tau + e_x$, where ω_τ is the true tax wedge. The OLS estimate of $L_{x\tau}$ from the projection of $\ln x$ on $\ln \omega_{\tau^*}$ is: $\hat{L}^{OLS}_{x\tau} = L_{x\tau} \times \frac{\sigma^2_\tau}{\sigma^2_{\tau^*}}$.

$$\begin{aligned}
\ln(Z^*/Z)|\omega^*_\tau &= \frac{\alpha(1-\beta)}{1-\gamma}\frac{\sigma^2_{\tau^*}}{2} + \frac{\alpha(1-\beta)}{1-\gamma}\sigma^2_{\tau^*}(\hat{L}^{OLS}_{k\tau} - 1) + \frac{\alpha\beta}{1-\gamma}\sigma^2_{\tau^*}\hat{L}^{OLS}_{n\tau} \\
&= -\frac{\sigma^2_\tau + \sigma^2_\epsilon}{2}\frac{\alpha(1-\beta)}{1-\gamma} + \frac{\alpha(1-\beta)}{1-\gamma}\sigma^2_\tau L_{k\tau} + \frac{\alpha\beta}{1-\gamma}\sigma^2_\tau L_{n\tau} \\
&= \ln(Z^*/Z)|\omega_\tau - \frac{\sigma^2_\epsilon}{2}\frac{\alpha(1-\beta)}{1-\gamma}.
\end{aligned}$$

□

A.5 Computation of the non-linear gains and measurement error correction

For each $x \in \{\omega_K,\ \omega_N,\ z\}$, define $\ln \hat{x} = b_{x\tau} \times \ln \omega_\tau$, where $b_{x\tau}$ is the OLS estimator of $L_{x\tau}$. We ignore the intercept term in that projection, because it cancels out from the TFP equations as it is common to all firms. Note also that we are using the OLS estimator, not the consistent estimator.[A1]

We then compute current TFP by substituting $\hat{\omega}_K$, $\hat{\omega}_N$ and $\hat{z}$ in equation (4), and the uniform-tax TFP by substituting $\hat{\omega}_N$, $\hat{z}$, and $\omega'_k = \hat{\omega}_K/\omega_{\tau^*}$ for ω_K in the same equation. This yields the following equations:

[A1] Because various biases from using the OLS estimator cancel each other out, the resulting formula for error correction is simpler and aligns with that reported in Proposition 4. We also considered using the consistent (IV) estimator, and a non-parametric estimator for $\mathbb{E}[\ln x|\omega_\tau]$. Both yield similar results.

$$\hat{Z}_{\tau^*} = \frac{\int (\omega_{\tau^*})^{\frac{1}{1-\gamma}(b_{z\tau}-\beta b_{n\tau}-\alpha b_{k\tau})} dG_{\tau^*}}{\left[\int (\omega_{\tau^*})^{\frac{1}{1-\gamma}(b_{z\tau}-\beta b_{n\tau}-(1-\beta) b_{k\tau})} dG_{\tau^*}\right]^{\alpha} \left[\int (\omega_{\tau^*})^{\frac{1}{1-\gamma}(b_{z\tau}-(1-\alpha) b_{n\tau}-\alpha b_{k\tau})} dG_{\tau^*}\right]^{\beta}}$$

$$\hat{Z}^*_{\tau^*} = \frac{\int (\omega_{\tau^*})^{\frac{1}{1-\gamma}(b_{z\tau}-\beta b_{n\tau}-\alpha b_{R\tau})} dG_{\tau^*}}{\left[\int (\omega_{\tau^*})^{\frac{1}{1-\gamma}(b_{z\tau}-\beta b_{n\tau}-(1-\beta) b_{R\tau})} dG_{\tau^*}\right]^{\alpha} \left[\int (\omega_{\tau^*})^{\frac{1}{1-\gamma}(b_{z\tau}-(1-\alpha) b_{n\tau}-\alpha b_{R\tau})} dG_{\tau^*}\right]^{\beta}},$$

where $b_{R\tau} = b_{k\tau} - 1$ is the projection coefficient of $\ln \omega_R$ on $\ln \omega_{\tau^*}$ and G_{τ^*} is the marginal distribution of the measured tax wedge.

When ϵ is normally distributed with mean zero and variance σ_ϵ^2 and is independent of ω_τ, then, for any scalar c, $E[(\omega_{\tau^*})^c] = E[\omega_\tau^c e^{c\epsilon}] = E[\omega_\tau^c] \cdot E[e^{c\epsilon}] = E[\omega_\tau^c] \cdot \exp(c^2\sigma_\epsilon^2/2)$. Replacing c with the appropriate power in each term gives:

$$\ln \frac{\hat{Z}^*_{\tau^*}}{\hat{Z}_{\tau^*}} = \ln \frac{\hat{Z}^*_{\tau}}{\hat{Z}_{\tau}} - \frac{\sigma_\epsilon^2}{2} \frac{\alpha(1-\beta)}{1-\gamma} \tag{A21}$$

Repeating these steps for the alternative formulations in Online Appendix A.2 by substituting τ^* for τ in the expressions for $\hat{Z}^{**}_{\tau^*}$ and $\hat{Z}^{***}_{\tau^*}$ presented in equations (OA.2) and (OA.6) give the following relations between the measured TFP loss and the actual TFP loss:

$$\ln \frac{\hat{Z}^{**}_{\tau^*}}{\hat{Z}_{\tau^*}} = \ln \frac{\hat{Z}^{**}_{\tau}}{\hat{Z}_{\tau}} - \frac{\sigma_\epsilon^2}{2} \frac{\gamma}{1-\gamma} \tag{A22}$$

$$\ln \frac{\hat{Z}^{***}_{\tau^*}}{\hat{Z}_{\tau^*}} = \ln \frac{\hat{Z}^{***}_{\tau}}{\hat{Z}_{\tau}} \times \frac{\sigma_\tau^2}{\sigma_\epsilon^2 + \sigma_\tau^2} \tag{A23}$$

B Data Appendix

The firm-level dataset used in Section 3 was constructed as follows. We use the annual Compustat database (Standard & Poor's, 2023), which provides balance-sheet data on publicly listed companies in the U.S. Our sample includes the years 1980–2021. We restrict attention to firms registered

in the U.S. and exclude firms in the finance, insurance, and real estate sectors, as well as in utilities and public administration. We remove observations with negative sales.

We construct firm-level capital stocks using a perpetual inventory method. For each firm, we initialize the capital stock in the first year for which both gross and net property, plant, and equipment (PPEGT and PPENT) are available, using PPEGT deflated by the investment price deflator. In subsequent years, we update the capital stock by adding the change in PPENT, deflated by the average investment price deflator over the corresponding endpoints. Output y_{it} is the sum of sales and changes in inventories. Firm-level TFP is computed as $\log z_{it} = \log y_{it} - \beta \cdot \log n_{it} - \alpha \cdot \log k_{it}$, where factor shares come from the BLS and are computed for the year 2000, the midpoint of our sample. We remove NAICS-year fixed effects from firm-level TFP.

We supplement these data with information on firms' marginal tax rates taken from *i)* Graham and Mills (2008), abbreviated as "GM" and *ii)* Blouin et al. (2010), abbreviated as "BCG".[A2] While the GM data cover the years 1980–2021, the BCG data are only available from 1980–2016. We link the Compustat data to the marginal tax rate data via the firm identifier GVKEY. Finally, we remove firm-year observations for which both tax rates are missing.

This results in a sample of 185,203 unique firm-year observations, averaging about 4,400 firms per year. Effective marginal tax rates are available for 70.3 percent of our observations (90.2 percent for the BCG tax rates).

Table A1 summarizes the two EMTR measures. For comparison, the top federal corporate income tax rate changed from 46 percent in 1980 to 35 percent from 1993 through 2017, and 21 percent thereafter. Before 2018, the corporate tax schedule was graduated, so marginal statutory rates could also vary with taxable income. Rates estimated by Graham and Mills (2008) show more bunching at zero and at the top statutory rate. Rates estimated by Blouin et al. (2010) display a smoother distribution, with a higher average rate and a slightly smaller variance. The two measures are highly but imperfectly correlated ($\rho = 0.62$).

[A2]Each study contains two measures of the marginal tax rates: before and after interest deductions are applied. The marginal tax rates we use in this paper are after interest deductions.

Combining estimates across tax measures The TFP calculations require a value of each projection coefficient for every year. Equations (9) and (10) produce two IV estimates, obtained by alternately using the GM and BCG measures. We combine the two estimates year by year using inverse-variance weights. For $x \in \{k, n\}$, let $\hat{L}_{x\tau}^{\mathrm{GM}}$ and $\hat{L}_{x\tau}^{\mathrm{BCG}}$ denote the estimates obtained using the two tax measures, with estimated variances V_{GM} and V_{BCG}. We combine them using inverse-variance weights:

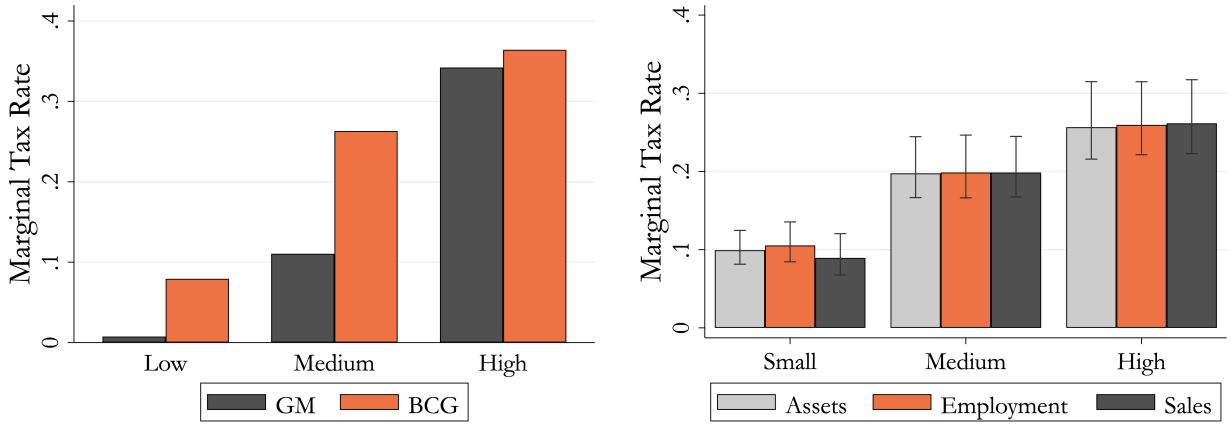
$$\hat{L}_{x\tau} = \frac{V_{\mathrm{BCG}}}{V_{\mathrm{GM}} + V_{\mathrm{BCG}}} \hat{L}_{x\tau}^{\mathrm{GM}} + \frac{V_{\mathrm{GM}}}{V_{\mathrm{GM}} + V_{\mathrm{BCG}}} \hat{L}_{x\tau}^{\mathrm{BCG}}.$$

If the two estimates are uncorrelated, the variance of the combined estimate is

$$\mathrm{Var}(\hat{L}_{x\tau}) = \frac{V_{\mathrm{GM}} V_{\mathrm{BCG}}}{V_{\mathrm{GM}} + V_{\mathrm{BCG}}}.$$

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(a) Tax rates by tax tercile

(b) Tax rates by size class

Figure 1: Asymmetry in Effective Marginal Corporate Tax Rates

Note.– GM and BCG refer to the rates estimated by Graham and Mills (2008) and Blouin et al. (2010). The figure shows average marginal corporate tax rates by tax-rate tercile (panel a), and by terciles of firm size as measured by assets, employment, or sales (panel b). The high and low values shown with whiskers correspond, respectively, to the two tax-rate estimates by GM and BCG. Bars show their average. Source: Compustat. The sample covers 1980–2021; GM rates are available through 2021 and BCG rates through 2016.

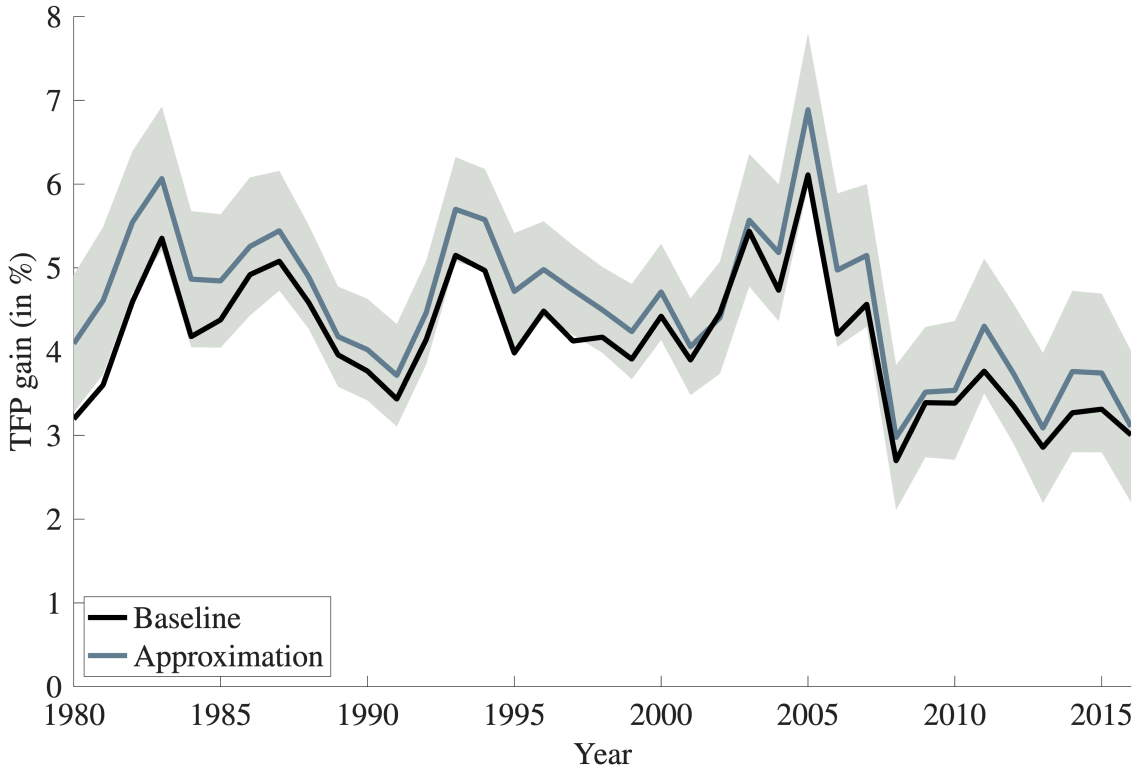


Figure 2: TFP gains from tax rate equalization

Note.– The figure shows the estimated change in aggregate TFP if all firms were to face a common corporate tax rate in the baseline model and its approximation using equation (7).

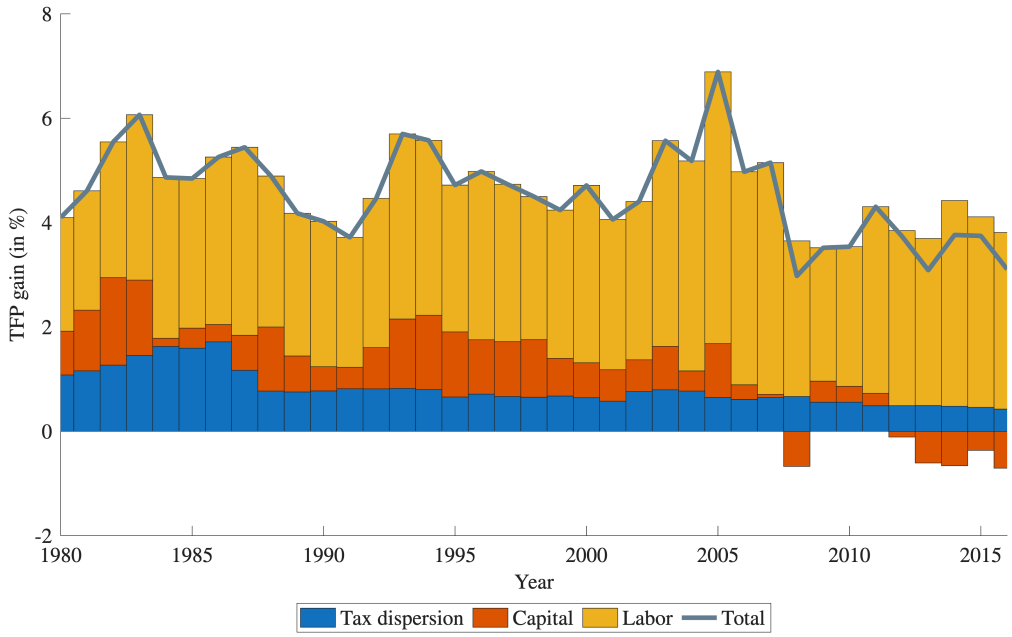


Figure 3: Components of TFP gains from eliminating tax heterogeneity

Note.– The figure shows the three components of TFP gains from tax rate equalization across firms (see equation (7)). The blue bars labeled "Tax dispersion" represent gains in an otherwise frictionless economy. The red bars labeled "Capital" show additional gains/losses in an economy with capital distortions. The bars labeled "Labor" show additional gains in an economy with capital and labor distortions.

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Table 1: Tax distortions and factor productivity

	(1) $\ln y/k$	(2) $\ln y/n$	(3) $\ln z$	(4) $\ln y/k$	(5) $\ln y/n$	(6) $\ln z$
			Panel A			
$\ln \omega_{\mathrm{GM},\tau}$	0.67	0.55	0.56	1.26	1.61	1.41
	(0.04)	(0.03)	(0.03)	(0.08)	(0.06)	(0.05)
			Panel B			
$\ln \omega_{\mathrm{BCG},\tau}$	0.96	1.23	1.07	1.59	1.31	1.33
	(0.06)	(0.04)	(0.04)	(0.09)	(0.06)	(0.06)

Note.– The table shows the results from regressions of firm-level capital, labor, and TFP on the log tax wedge, $\ln[1/(1-\tau)]$. Columns 1 to 3 report OLS estimates. Columns 4 to 6 report instrumental variable estimates to correct for measurement error. All specifications control for a full interaction of sector and year indicators. Standard errors are clustered at the firm level. The first-stage F-statistics for the IV regressions in Columns 4-6 range from 46,665 to 49,592. Data on productivity come from authors' calculations from Compustat. Data on marginal tax rates come from Graham and Mills (2008) in Panel A and Blouin et al. (2010) in Panel B.

	Total	Capital-Linked	Labor-Linked	Tax Alone
Benchmark	4.6	0.6	3.2	0.8
Structural Distortions				
Capital	5.7	0.8	4.2	0.8
Labor	10.7	1.3	3.8	5.6
Declining labor share and				
rising capital share	4.7	0.6	3.3	0.9
rising markup	4.0	0.5	2.7	0.8

Table 2: TFP Gains from Homogenizing Tax Rates: Extensions and Robustness

Note.– Percent change in aggregate TFP if all firms were to face a common corporate tax rate. Average estimates over the 1980–2016 period. Baseline model assumes corporate taxes only distort investment. The middle panel allows taxes to structurally affect pretax capital or labor costs. Bottom panel allows for declining labor share of income compensated by either a higher capital share or higher price markups.

Table A1: Summary statistics of marginal tax rates

Variable	mean	sd.	p25	p50	p75	N
τ^{GM}	0.168	0.171	0.007	0.062	0.341	130,232
τ^{BCG}	0.236	0.138	0.101	0.277	0.341	167,014

Note.– The two effective tax variables are taken from Graham and Mills (2008) and Blouin et al. (2010).